AJOKU MIRACLE VICTORY

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TECHNICAL SKILLS

- Process Simulation: Aspen Plus (custom unit operations via Fortran subroutines, multicomponent systems), Aspen HYSYS (advanced: carbon capture, amine sweetening), COMSOL Multiphysics (heat/mass transfer, fluid dynamics), DWSIM, Design Expert, SolidWorks (basic), Atomic Simulation Environment (ASE)
- Computational Modelling & ML: Python (NumPy, Pandas, Scikit-learn), Machine Learning, Artificial Neural Networks (ANN), Surrogate/Proxy Modelling, Physics-Informed ML, Numerical Optimization (SQP), Atomic Simulation Environment (ASE), MATLAB/Simulink, LLMs, COMSOL Multiphysics, Multi-objective optimization (pymoo, NSGA-II)
- Data Science & Analytics: Data Analysis & Visualization, SQL, Power BI, Tableau, Advanced Excel, Streamlit,
- Core Engineering: Mass & Energy Balance, Thermodynamic Optimization, Techno-Economic Analysis (TEA), Process Flow Diagram (PFD) Development, Numerical Analysis, Statistical Analysis, Technical Writing
- Professional: Technical Leadership, Project Management, Interdisciplinary Collaboration, Curriculum Development, Scientific Communication

EDUCATION

Federal University of Technology Owerri, Nigeria — *November 2018 – July 2024*
BEng, Chemical Engineering | CGPA: **4.42/5.00** (SCHOLARO GPA: **3.71/4.00**) — Class Rank: 6/176

- Awards: Federal Scholarship Board Merit Scholar; PTDF Merit Scholar

PUBLICATIONS & MANUSCRIPTS

Published

- Anyikwa S. O., **Ajoku M. V.**, *et al.* (2026). The potential of amine-to-feed flowrate ratio for post-combustion carbon capture systems using Aspen HYSYS. *Journal of Business, IT, and Social Science*, 5(1), 97–103. https://doi.org/10.51470/bits.2026.05.01.97 ***[Co-lead author]***

Preprints/Submitted

- **Ajoku M. V.**, Sola-Ojo M., Duru G. C. (2026). "Can pure-gas permeability data design a mixed-gas module? A correctability map for dual-mode CO_2/N_2 transport in Matrimid 5218 hollow fibres". *Research Square* (preprint): https://doi.org/10.21203/rs.3.rs-11646086/v1. Submitted to *Theoretical Foundations of Chemical Engineering*. Code and data: https://doi.org/10.5281/zenodo.23070727
- **Ajoku M. V.** (2026). "Membrane pre-concentration ahead of MEA absorption: a rate-based, mixed-gas techno-economic assessment for natural gas, coal and cement flue gases". *Research Square* (preprint): https://doi.org/10.21203/rs.3.rs-11674132/v1. Submitted to *Chemical and Biochemical Engineering Quarterly*. Code: https://doi.org/10.5281/zenodo.23263315
- **Ajoku M. V.**, Ezeokolie E. D., Nweke B. C., Taofeeq M. A., Ojinika A. P., Akoh M. D., Tile S. T. (2026). "Design, Construction and Performance Evaluation of a Low-Cost Pilot-Scale Semi-Batch Pyrolysis Plant for Converting Polystyrene and Low-Density Polyethylene Waste into Fuel Oil". *Research Square* (preprint): https://doi.org/10.21203/rs.3.rs-11337565/v1 Submitted to *Brazilian Journal of Chemical Engineering*
- Owuala S. O., **Ajoku M. V.**, Asooye P. A., Ogunseye O. M., Mogaji D. C., Ikemdinachi B. I. "Prompt Injection Attacks in Low-Code AI Platforms: A Threat Taxonomy and Mitigation Strategy". Submitted to *Acta Informatica Pragensia*. Code and data: https://doi.org/10.5281/zenodo.23449065

Manuscripts Under Review

- **Ajoku M. V.**, Odigie T. O., Ezeokolie E. D., et al. "Benchmarking Machine Learning Models for Offshore Oil Production Forecasting: A Multi-Horizon, Multi-Architecture Study on the VOLVE Dataset". Under review at *Archives of Advanced Engineering Science*.
- **Ajoku M. V.** et al. "Methane Persistence, Public Accountability, and Environmental Justice in the New Mexico Permian Basin". Under review at *Journal of Environmental Justice*. Code and data: https://doi.org/10.5281/zenodo.22739020
- Owuala S. O., Sidiqot T. O., Adedero C., Chidera S. I., Emmanuel U. A., **Ajoku M. V.** "Prompt personalization at scale: A framework for adaptive task assistants using LLMs". Under review at *Journal of Artificial Intelligence and Robotics Research*.

PEER REVIEW

- **Reviewer,** *Petroleum Science* (China University of Petroleum; published by KeAi / Elsevier), 2026 (1 manuscript)

RESEARCH EXPERIENCE

Undergraduate Researcher, Federal University of Technology Owerri — Imo State, Nigeria
Carbon Capture Systems: A Study using Aspen Hysys — *August 2023 – July 2024*
Research Supervisors: Engr. Prof. O. E. Onyelucheya, Engr. S. C. Anyikwa

- Modelled and optimized an amine-based CO_2 absorption–stripping process in Aspen HYSYS through coupled mass and heat transfer analysis, identifying optimal operating conditions (absorber pressure: 220 kPa; solvent-to-feed ratio: 7.0; separator temperature: 25 °C) achieving ~100% capture efficiency.
- Quantified a 12–15% reduction in regenerator energy demand through parametric optimization of regenerator feed temperature, establishing clear thermodynamic trade-offs without compromising capture performance.

Undergraduate Researcher, Federal University of Technology Owerri — Imo State, Nigeria
Acetic Anhydride Plant Design using the Cativa Process Route — *August 2023 - May 2024*
Research Supervisors: Engr. Prof. M.S. Nwakaudu, Engr. S. C. Anyikwa, Engr. B.C. Ngolube

- Spearheaded a 15-member team in designing a 100,000 MTA acetic anhydride production plant utilizing Nigeria's Anambra Basin coal reserves via the low-energy Cativa process (iridium-catalyzed carbonylation).
- Executed process design calculations including mass/energy balances, equipment sizing (reactors, distillation columns), and utility requirements to ensure operational feasibility.
- Developed integrated process diagrams: Optimized flowsheets in Aspen Hysys v11 simulating reaction-separation units; Drafted P&ID and QFD using Microsoft Visio 2021 and protocols for safety, environmental compliance, and operations.

Collaborative Research/ Independent Projects
Proxy Model for Critical Rate (qoc) and Perforation Depth (hopt) — *December 2023 – March 2024*

- Developed Python-based proxy model predicting qoc and hopt from GOC data, achieving 2% relative error against Papatzacos' benchmark method.
- Utilized Python libraries including NumPy, Pandas, and Scikit-learn for data manipulation, model training, and validation.

ANN-Driven Waterflood Optimization — *November 2024 – January 2025*

- Co-led development of 3-layer ANNs (ReLU activation) correlating injection rates with production metrics.
- Pioneered formulation of [Liquid Rate - Water Cut] objective function to identify optimal parameters.

DATA SCIENCE & ANALYTICS PROJECTS

Accident Analysis using Machine Learning

- Developed an interactive Streamlit application to analyze road accident severity using UK traffic datasets. Applied classification algorithms to predict accident outcomes based on weather, time, and location variables. [***Streamlit App***][***GitHub Repo***]

Customer Segmentation using RFM & K-Means

- Performed RFM (Recency, Frequency, Monetary) analysis on over 500,000 transaction records to segment customers into 4 distinct groups using Python (Pandas, Seaborn, Scikit-learn).
- Applied K-Means clustering with elbow method and silhouette score to validate optimal clusters [***GitHub Repo***].

Supply Chain Analytics: On-Time Delivery & Cost Control (SQL + Power BI)

- Analyzed 11,000+ shipment records using MySQL and Python to identify on-time delivery drivers; built an interactive Power BI dashboard improving KPI tracking visibility by 15% for senior management [***GitHub Repo***].

State of Content Writing in Nigeria, 2025

- Collaborated on a comprehensive study analyzing trends, earnings, and challenges for 100+ Nigerian content writers. Performed data cleaning and analysis on survey data, generating actionable insights into industry growth and income distribution. [***Published Article***] |[***PDF Report***] [***GitHub Repo***].

WORK EXPERIENCE

Data Operations Analyst / Business Optimization
MEDLOG Transport and Logistics (Subsidiary of MSC), Lagos, Nigeria | September 2025 – Present

- Designed and deployed an automated reporting pipeline for fleet optimization, cutting manual reporting effort and improving turnaround for operational visibility.
- Analyzed container movement and logistics KPI data (SQL, Advanced Excel) and built Power BI dashboards giving management real-time view of throughput and turnaround.
- Led root-cause analysis of system and data discrepancies, implementing fixes that reduced recurring reporting errors.

Process Simulation Intern
Dumac Ibolo Services, Port Harcourt | May 2023 – August 2023

- Developed process simulations in Aspen Hysys/Aspen Plus, optimizing energy efficiency for industrial flue gas systems.
- Analyzed simulation data to identify process improvements, reducing regeneration energy by 8% in case studies.

Academic Tutor (STEM)
GauthExpert, Remote | March 2023 – January 2026

- Solved 1500+ advanced STEM problems (Calculus, Chemistry, Physics) for high school/early-college students under strict 30-50 minute time constraints.
- Delivered accurate solutions with conceptual explanations, maintaining 95%+ accuracy rate and 90%+ user satisfaction.

TEACHING EXPERIENCE

Lead Academic Tutor and Curriculum Developer
Redeemed Christian Fellowship, Federal University of Technology Owerri| November 2019 – April 2023

- Directed academic support programs for 300+ students across 4 cohorts, including course retakers.
- Designed and delivered weekly tutorials in **Engineering Mathematics**, **Reaction Engineering, Process Optimization, and Transport Phenomena.**
- Implemented exam-prep frameworks resulting in **100% pass rate** among participating retakers and 40% grade improvement average.

Freshman Academic Coordinator

Redeemed Christian Fellowship, Federal University of Technology Owerri | January 2019 – October 2019

- Spearheaded foundational STEM instruction for 200+ first-year students.

LEADERSHIP EXPERIENCE

Group Lead - Undergraduate Thesis

Federal University of Technology Owerri | August 2023-July 2024

- Directed 10-member research team in carbon capture simulations (Aspen Hysys), coordinating experimental design to final report.

Group Lead - Plant Design Project

Federal University of Technology Owerri | August 2023 - May 2024

- Led a 15-person team designing 100,000 MTA acetic anhydride plant (Cativa process).
- Streamlined research/P&ID development using collaborative tools (Visio, Teams), earning top 5% project grade

HONOURS AND AWARDS

Nigerian Universities Engineering Student Association (FUTO) Engineering paper of the Year | *2018/2019*

- Recognized as the top undergraduate engineering research paper of 2018/2019 for technical originality and scholarly contribution.

Petroleum Technology Development Fund (PTDF) merit Scholarship Award | 2022 - 2024

- Scholarship worth 800,000 naira per session awarded to top 1% of Nigerian engineering students based on academic excellence and research potential in energy related fields.

Federal Scholarship Board (FSB) merit Scholarship Award| 2020 - 2024

- Scholarship worth 250,000 naira per session to support undergraduate study; competitively selected among 5,000+ applicants nationwide for sustained academic distinction throughout undergraduate studies.

PROFESSIONAL AFFILIATIONS

- NSChE: Nigerian Society of Chemical Engineers; NUESA: Nigerian Universities Engineering Students Association

LANGUAGES

- English: Native / Fluent (professional and academic level); German (Basic)

COMMUNITY SERVICE AND VOLUNTEERING

Community Development Officer / Advocate | NYSC NDLEA Community Development Service | July 2025 – May 2026

- Spearheaded grass-roots public health campaigns and anti-drug advocacy programs across local government schools, markets, and community hubs.
- Collaborated with NDLEA command officials to design and distribute informative materials detailing the socio-economic impact of substance abuse.

Academic Development Strategist | Redeemed Christian Fellowship, FUTO | 2019 - 2023

- Spearheaded peer-to-peer academic intervention programs supporting 200+ struggling students.
- Mobilized volunteer tutors across 5 engineering departments for cross-disciplinary academic support.

REFERENCES

Engr. Prof. Okechukwu Onyelucheya
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Relationship: Academic Advisor/Lecturer

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